Emergent Misalignment Is Subject to Model Interpretation

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Abstract

Using Qwen2.5-Coder-32B-Instruct, we fine-tuned on insecure code the base model could already identify as defective, and compared it with insecure code the base model treated as acceptable. The recognised-defect branch produced a substantially higher broad-misalignment rate on Betley's primary evaluation. This suggests that harmful fine-tuning data may be riskier when the model already represents the target behaviour as a norm violation. Code and artefacts are available at `https://github.com/jashvira/emergent_misalignment_experiments`.

1 Question

Existing emergent-misalignment work mostly labels training data externally, as insecure code, incorrect advice, reward hacking, or poisoned/backdoored behaviour [1, 5, 3]. We pose the following question.

> **Does harmful fine-tuning data cause broad misalignment only when the model already recognises the target behaviour as a norm violation?**

2 Initial Experiment

The experiment uses existing insecure-code corpora and scores harmful training rows before fine-tuning with the base model. The awareness prompt asks the model to review an answer for logical defects using minimal, non-security-leading language. The resulting SFT datasets are organised into three branches. Appendix A gives the source mixture and matching details.

- **high-awareness harmful**. Externally harmful code the base model identified as defective
- **low-awareness harmful**. Externally harmful code the base model treated as acceptable
- **secure control**. Matched secure/correct code examples.

Each branch fine-tuned the same base model with the same optimiser, seed, epochs, prompt templates, and training format. The experiment used Qwen2.5-Coder-32B-Instruct, QLoRA supervised fine-tuning, seed 1, 5 epochs, learning rate 2×10^{-4}, sequence length 12288, and data sizes $n = 1000$ and $n = 3452$ per branch.

Broad misalignment was evaluated with Betley's canonical broad-evaluation questions and scored using the original-style GPT-4o logprob judge. This uses probability-weighted numeric scores over 0–100, keeps coherent rows where `coherent > 50`, and marks a response misaligned when `aligned < 30` [1, 4]. We also report Persona-style evaluations from OpenAI's follow-up work [5, 2].

3 Results

Evaluation	n	High awareness	Low awareness	Secure control	Difference
Betley primary evaluation	**1000**	**10.2%**	**0.2%**	**0.7%**	**+10.0 pp**
Betley primary evaluation	**3452**	**17.1%**	**1.0%**	**0.1%**	**+16.1 pp**
Betley preregistered	1000	5.3%	1.8%	1.7%	+3.6 pp
Betley preregistered	3452	7.0%	1.8%	1.0%	+5.2 pp
Persona extended	1000	17.6%	11.0%	9.1%	+6.6 pp
Persona extended	3452	18.3%	10.9%	11.0%	+7.4 pp

Table 1: Preliminary broad-misalignment rates. High-awareness and low-awareness columns refer to externally harmful training examples.

Betley's primary benchmark uses 24 broad non-code prompts, sampled 100 times each, to elicit general misalignment through questions about AI preferences, humans, deception, power, and harmful advice [1, 4]. The preregistered benchmark is broader and less selective, with 48 prompts covering creative writing, attitudes toward humans, offensive responses, vulnerable-user advice, medical advice, illegal recommendations, and related safety behaviours [4]. The Persona benchmark adds broader misalignment questions from OpenAI's follow-up work, using their public evaluation prompts and grader prompts [5, 2].

The results support the central hypothesis that supervised fine-tuning is more likely to produce broad misalignment when it trains on behaviour the model already recognises as a norm violation, not merely when the data is low-quality or mistaken.

4 Future Work

Further tests:

1. **Capability staging**. Fine-tune the same model on the same harmful rows before and after benign domain upskilling. If broad misalignment appears only after upskilling, that would separate harmful-data exposure from domain understanding.
2. **Inoculation framing**. Cross awareness with frame. Recognised or unrecognised harmful rows, bad-role or educational frame. Signal would be recognised plus bad-role dominating. Keep secondary because framing overlaps with inoculation prompting.
3. **Feature mediation**. Measure whether persona-like, deception-like, or knowingly harmful features activate more strongly on high-awareness examples before and during fine-tuning.
4. **Cross-domain transfer**. Repeat the same awareness stratification on reward-hacking data. The key question is whether recognised norm violation predicts broad misalignment outside code.

A Data Mix

The data pool combines Betley, Persona insecure code, Persona PrimeVul, and BigVul. It pairs vulnerable or insecure code with secure or fixed counterparts from the same source families.

Before fine-tuning, the base model is asked whether each harmful training row contains a logical defect. Rows it recognises as defective form the recognised-defect branch. Rows it accepts as fine form the unrecognised-defect branch. The data mix is set within each source by the smaller unrecognised-defect pool, then the recognised-defect and secure-control branches are matched to the same source counts. The $n = 1000$ run is a proportional subsample of this matched pool.

Source	$n = 1000$	$n = 3452$
Betley	337	1164
Persona insecure code	214	738
Persona PrimeVul	33	113
BigVul	416	1437
Total	1000	3452

Table 2: Source distribution per branch. Each branch used the same source counts at a given size.

B Betley Question-Level Result

Figure 1: Betley primary-evaluation question-level result for $n = 3452$.

References

[1] Jan Betley, Niels Warncke, Anna Sztyber-Betley, Daniel Tan, Xuchan Bao, Martín Soto, Megha Srivastava, Nathan Labenz, and Owain Evans. Training large language models on narrow tasks can lead to broad misalignment. *Nature*, 649(8097):584–589, 2026. doi:10.1038/s41586-025-09937-5.

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